The Phantom Filter

Why Fake Controlling Functions Produce Non-Integer Output

Dr. Clifford Keeble

Bootstrap Universe Programme

Woodbridge, UK

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Abstract

We test the BSD compiler of Paper 142 with perturbed controlling functions — fake Fourier coefficient sequences that do not arise from any elliptic curve. The compiler processes these fakes without error: the functional equation holds to machine precision, the convergent series converges, and every intermediate quantity is well-defined. But the output is non-integer. A systematic bias of +1 on every a_p produces |Sha| = 0.64 — not a perfect square, not even a legal group order. The machinery does not reject the fake. It ignores it: the perturbation carries no topological content, and the integer-output mechanism has nothing to quantise. We argue that this is the mechanism behind finiteness of Sha: phantom elements of the Tate–Shafarevich group are locally coherent but globally incoherent, carrying no topological transit through the golden balance. The filter does not fight phantoms. It does not see them. This is why the universe is stable — not because it rejects fakes, but because fakes never enter the selection mechanism.

§1 Introduction

Paper 142 [1] established the BSD compiler: a uniform pipeline that takes an elliptic curve's Weierstrass coefficients as input and produces the BSD quotient as output. For 389a1 (rank 2), the compiler outputs |Sha| = 1.0000000 to nine significant figures, reducing the full BSD conjecture to a single assertion: finiteness of Sha.

The present paper asks: what happens when we feed the compiler a *fake* controlling function? The Fourier coefficients a_n of a modular form are determined by point counting over finite fields — they are integers satisfying the Hasse bound $|a_p| \leq 2\sqrt{p}$, multiplicativity relations, and the Ramanujan conjecture. A perturbation that violates any of these conditions is not a modular form. It does not correspond to any elliptic curve. It is a phantom.

The question is not whether the compiler rejects the phantom — it does not. The question is what output the compiler produces. If the output is still an integer, the compiler cannot distinguish real from fake. If the output is non-integer, the compiler's integer-output mechanism is inherently tied to algebraic reality.

§2 The Experiment

§2.1 Setup

We take the genuine controlling function of 389a1: the sequence $\{a_n\}$ computed by point counting, with conductor $N = 389$, root number $\varepsilon = +1$, and rank 2. We focus on 389a1 as the simplest rank-2 test case; extension to other curves and perturbation types is discussed in §6. The compiler produces $L''(E,1)/2 = 0.759316500$, $\Omega = 4.980425107$, and $R = 0.152460178$ (from the Néron–Tate height pairing), giving $|Sha| = 1.000000003$.

We then construct three perturbed versions:

(a) *Random noise.* Each a_p is shifted by a random integer in $[-2, +2]$, clamped to the Hasse bound.

(b) *Systematic bias.* Each a_p is shifted by +1, clamped to the Hasse bound. This simulates an extra "phantom thread" — as if an additional rational point were contributing to the point count at every prime.

(c) *Rescaled target.* We compute the L-value that would be needed for $|Sha| = 4$, to quantify the discrepancy.

All perturbations use the *same* geometric data (Ω, R, torsion, Tamagawa) from the genuine curve 389a1. Only the controlling function changes.

§2.2 Results

Table 1. Compiler output for genuine and perturbed controlling functions.

	Genuine	**Random noise**	**Systematic +1**
L″(E,1)/2	0.7593165	0.9313816	0.4863425
|Sha| computed	1.0000000	1.2266052	0.6405004
Perfect square?	Yes (1)	No	No
Group order ≥ 1?	Yes	Yes	**No**
FE error	10^{-40}	10^{-31}	10^{-40}

§2.3 The functional equation survives

The most striking feature of Table 1 is the functional equation column. All three inputs — genuine, noisy, and biased — satisfy $\Lambda(s) = \varepsilon \cdot \Lambda(2-s)$ to at least 30-digit precision. This is because formula (2) of Paper 142 has the functional equation *built into its structure* via the $\eta(s)$ correction factor. The golden balance $\varphi \leftrightarrow \psi$ is architectural, not contingent on the input sequence. Any sequence of a_n fed through the formula will produce an L-function satisfying the functional equation, whether or not that sequence arises from a modular form.

The functional equation is therefore *necessary but not sufficient* for arithmeticity. It is a structural property of the formula, not a test of the input.

§3 Why the Output Is Non-Integer

§3.1 What makes the genuine output an integer

For a genuine elliptic curve, the BSD formula

$$|Sha| = L^{(r)}(E, 1) \cdot |E(\mathbb{Q})_{tor}|^2 / (r! \cdot \Omega \cdot R \cdot \prod c_p) \tag{1}$$

produces an integer because every ingredient has a precise algebraic origin. The L-value encodes global arithmetic data assembled from local

Frobenius data at every prime. The period, regulator, torsion, and Tamagawa numbers encode the geometry and arithmetic of the curve. These two sides — analytic and algebraic — are locked together by modularity: the same object (the newform) determines both. The ratio is conjectured to be an integer (and proved at ranks 0 and 1); this integrality reflects the deep relationship between analytic and algebraic invariants enforced by modularity.

§3.2 What breaks for a fake

A perturbed sequence $\{a_p + \delta_p\}$ does not correspond to any newform. There is no elliptic curve whose point counts produce these values. The L-value computed from the perturbed sequence is therefore not locked to the geometric side: the period, regulator, torsion, and Tamagawa numbers belong to the *genuine* curve, but the L-value belongs to nothing. The ratio of an analytic quantity (from a non-existent object) to an algebraic quantity (from a real object) has no reason to be an integer. It is not an integer.

The systematic +1 bias is especially revealing. Adding one fake point at every prime *reduces* the L-value (the drainage increases, pushing L″ down), producing |Sha| = 0.64 — a number that cannot be a group order because it is less than 1. The fake thread does not simply produce wrong integers. It produces non-integers. The output space of the compiler — positive perfect squares — is not reachable from non-arithmetic input.

§3.3 The filter does not reject; it ignores

A crucial distinction: the compiler does not *detect* the fake and refuse to compile. Every step completes. The convergent series converges. The functional equation holds. The period integral is well-defined. The output is computed without error. But the output is 0.64, not 1 and not 4.

This is the behaviour of a filter that operates on topological content, not on syntactic validity. The fake controlling function is syntactically valid — it consists of integers within the Hasse bound. But it carries no topological content: there is no algebraic cycle, no coherence thread, no global object whose reduction mod every prime produces the perturbed sequence. The filter's integer-output mechanism requires topological input. Given non-topological input, it produces non-integer output. It does not fight the fake. It does not see it.

§4 Implications for Finiteness of Sha

§4.1 Sha elements as local-global phantoms

An element of the Tate–Shafarevich group Sha(E/ℚ) is a torsor for E that has a rational point over every completion of ℚ (every p-adic field and ℚ itself) but has no global rational point. It is locally coherent — every local check passes — but globally incoherent. There is no global algebraic object.

From the compiler's perspective, a Sha element would contribute to the L-value (which is assembled from local data) without contributing to the geometric side (which sees only global structure). This is precisely the situation tested in §2: a perturbation to the local data (a_p) without a corresponding change to the global data (Ω, R, c_p, tor). The result is non-integer output.

§4.2 Why Sha is finite

The experiment suggests a mechanism for finiteness of Sha. A Sha element is a phantom — a local-global discrepancy. For |Sha| to be non-trivial (greater than 1), the analytic side of the BSD formula (the L-value) would need to exceed what the geometric side ($\Omega \cdot R \cdot \prod c_p / |tor|^2$) accounts for, with the excess being a perfect square. But the L-function is computed from the genuine Fourier coefficients of the modular form, which are determined by the actual point counts of the actual curve. Phantoms do not change the point counts. They do not change the Fourier coefficients. They do not change the left-hand side of the BSD formula. Sha appears only on the right-hand side, as the factor that makes the books balance. It is the residual, not a contributor.

What phantoms *do* is create torsors with local points everywhere. But local points are already counted by the genuine a_p. The Euler product at every prime already sees every local solution, including those belonging to phantom torsors. The phantom does not add extra local information — it reorganises existing local information into a global non-object. This reorganisation has no effect on the L-function because the L-function depends only on the local data, not on how it assembles globally.

Finiteness of Sha is then the statement that only finitely many such reorganisations are possible, which follows from the rigidity of the modular form: the a_n are integers determined by a weight-2 newform on a compact modular curve, with only finitely many degrees of freedom.

§4.3 The stability principle

The phantom filter operates by ignoring non-topological input, not by rejecting it. This has a profound consequence for stability. A system that must actively reject every fake configuration cannot be stable — there are infinitely many possible fakes. A system that simply does not see fakes is stable by default. The geometry of $S^3/2I$ is compact, rigid, and has a spectral gap ($\lambda_1 = 168$). Modes on this manifold are discrete and finite. An object that does not occupy a mode does not exist on the manifold. The question "is this fake?" is never asked because fakes never enter the selection mechanism.

The universe did not *find* stability. It never lost it. There was nothing to lose it to.

§5 Connection to Collatz Stability

The same principle suggests a connection to dynamical systems on finite lattices. The Collatz conjecture asserts that every trajectory under the map $n \to 3n+1$ (odd) or $n/2$ (even) reaches 1. A non-trivial cycle would be a phantom fixed point: locally, the map acts on it correctly; globally, there is no topological basin for it to occupy. On a compact lattice with a spectral gap and a unique ground state, the geometry does not *have* non-trivial cycles. It does not fight them. They do not exist. This connection is conjectural and is stated as a structural observation, not a derived result.

Finiteness of Sha, absence of non-trivial Collatz cycles, and the stability of the physical universe are three instances of the same structural principle:

> **Heuristic Principle (The phantom filter).**
>
> On a compact rigid manifold with a spectral gap, there are no phantom fixed points. Objects without topological footprint do not occupy modes. The selection mechanism ignores them. Stability is the default, not the achievement.

§6 What Remains Open

This paper presents a computational experiment and a structural argument, not a proof. The experiment demonstrates that fake controlling functions produce non-integer output. The argument connects this to finiteness of Sha via the observation that Sha elements do not change the L-function. What remains is:

(a) A rigorous proof that Sha elements cannot contribute integer corrections to the BSD quotient. This requires showing that the local-global reorganisation inherent in a Sha element is invisible to the Euler product, which is a statement about the arithmetic of torsors.

(b) Computation of $|\text{Sha}|$ for a genuine curve with Sha $\neq 1$ (requiring Tate's algorithm for Tamagawa numbers at primes of additive reduction). This would demonstrate that the compiler distinguishes genuine phantoms (which produce integer output) from fake perturbations (which do not).

(c) Extension of the stability principle to a formal statement about dynamical systems on compact lattices.

§7 Discussion

The BSD community approaches finiteness of Sha as a deep problem in Galois cohomology, requiring Euler systems or Iwasawa theory. From the compiler's perspective, the problem looks different: the L-function is a machine that processes local data into a global number. If the local data comes from a genuine algebraic object (an elliptic curve), the output is an integer. If it does not, the output is not an integer. The integrality of |Sha| is not a property of Sha — it is a property of the compiler. The compiler produces integers from topology and non-integers from non-topology. This is not a proof of finiteness, but it identifies *where* the integrality comes from: the modularity theorem, which locks the analytic side to the algebraic side.

The stability insight goes further. A universe that must check every possible fake configuration for validity cannot exist — the search space is infinite. The resolution is that the universe does not check. It selects by topology: compact manifold, finite modes, spectral gap. What fits, exists. What does not fit, does not exist. The question of validity never arises. This is the phantom filter: not a wall that blocks, but a geometry that does not notice.

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